

Summary

AI Robotics Engineer at AMTRAN TECHNOLOGY CO., LTD. with an M.S. in Electrical Engineering. Build robot control and learning systems across ROS 2, TM5S, NVIDIA Isaac Sim / Isaac Lab, Movelt 2, cuMotion, and Isaac GR00T, integrating perception, data workflows, policy inference, and real-robot validation.

Experience

- AMTRAN TECHNOLOGY CO., LTD. | AI Robotics EngineerMay 2025 - Present | New Taipei, Taiwan
- Built ROS 2 control and simulation environments connecting TM5S, NVIDIA Isaac Sim / Isaac Lab, Movelt 2, and cuMotion for motion planning, collision handling, pick-and-place, and cable tasks.
  - Delivered Isaac GR00T N1.7 data conversion, model training, policy inference, and TM5S real-robot testing, integrating camera data, robot state / action, policy services, and robot commands.
  - Integrated RealSense, YOLO OBB, AprilTag, and OpenCV for 3D pose estimation, hand-eye calibration, TF conversion, and visual servoing in physical robot workflows.
  - Built Quest 2 teleoperation and imitation-learning recording workflows; automated task execution and data collection reached 3-4x manual operation in the same time window.
  - Independently delivered robotics software from requirements and architecture through Docker deployment, testing, on-site integration, and maintenance; cable-insertion success improved from 70% to 90% in a 10-trial test.
- ISCOM | Software Engineer InternFeb 2022 - Jun 2022 | Kaohsiung, Taiwan
- Participated in backend API development, frontend integration, and web feature implementation using C#, ASP.NET MVC, JavaScript, jQuery, MS SQL, and Git.

Selected Projects

|                                           |                                                                                                                                                        |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Robot Workflow & Data Collection Platform | Next.js, React Flow, FastAPI; built a 34-node workflow system that raised data-collection throughput to 3-4x manual operation.                         |
| NVIDIA Isaac GR00T N1.7 VLA Integration   | Delivered data conversion, model training, policy inference, and TM5S real-robot testing across camera data, robot state/action, and control commands. |
| TM5S Visual Servoing & Cable Insertion    | Integrated RealSense, YOLO OBB, AprilTag, and OpenCV; improved insertion success from 70% to 90% in a 10-trial test.                                   |

Skills

|                            |                                                                                                                             |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Robot Control & Simulation | ROS 2, TM5S, NVIDIA Isaac Sim / Lab, Movelt 2, cuMotion, Docker, Linux                                                      |
| VLA & Robot Learning       | Isaac GR00T N1.7, VLA / Policy Inference, Imitation Learning, HDF5, Robot State / Action, Quest 2                           |
| Perception & Integration   | RealSense, OpenCV, YOLO OBB, AprilTag, Hand-Eye Calibration, Visual Servoing, Python, C++, FastAPI, React / TypeScript, Git |

Education & Honors

- M.S. in Electrical Engineering, National Taiwan Normal University | Sep 2022 - Jul 2024
- B.S. in Computer and Communication, National Pingtung University | Sep 2018 - Jun 2022
- Honors: RHCSA; FIRA Roboworld Cup First Place; National Technical Creativity Competition Third Place